COMMERCIAL ITEM DESCRIPTION

TAPE RETROREFLECTIVE, AND SHEETING RETROREFLECTIVE (PAVEMENT LINE, WORD AND SYMBOL MARKINGS)

The General Services Administration has authorized the use of this commercial item description.

This commercial item descripton covers retroreflective sheeting and tape to be used as pavement lines, words and symbols for traffic guidance.

The retroreflective product shall be of the following types and classes:

Type I - Reflective tape
Type II - Reflective sheeting

Class 1 - Pre-coated (pressure sensitive adhesive)
Class 2 - Applied with contact cement

Salient characteristics:

Thickness. The product thickness, without adhesive, shall not be less than 60 mils (0.06 inches).

Reflectance. A 2.0 x 2.5 ft. rectangular sample of the product shall have the following initial minimum reflectance values at an 86.0° entrance angle using Federal Test Method 370.

The test distance shall be 50 ft. and the angular aperture of both the photoreceptor and light projector shall be 6 minutes of arc. The reference center shall be the geometric center of the sample, and the reference axis shall be perpendicular to the test sample.

	WHITE		YELLOW	
Observation Angle	0.2°	0.5°	0.2°	0.5°
Specific Luminance as millicandelas per square foot per foot candle	550	380	410	250

Adhesive sheet strength. The average load required to break the adhesive bond of the product shall not be less than 3.2 kg (7 lb) when tested in the following manner:

The test shall be done in accordance with ASTM D 638 as modified to test the adhesive sheet strength. The test specimen shall be cut to dimensions of 1" x 6", a 3" piece of carbordundum extra coarse emory cloth, or its equivalent, shall be applied to the adhesive face of the test strip, so that there is a 1 in^2 overlap. A pressure of 50 psi shall be applied over this area for a period of 120 seconds. Load is applied by gripping each end of the test piece in a suitable tensile test instrument such as a Dillon or Scott tester. The speed of testing shall be 0.25"/min. The test shall be run in triplicate and at 25°C (77°F).

Color. The product shall be white or yellow conforming to the standard highway colors issued by the Federal Highway Administration.

Width. Type I reflective tape shall be a maximum of 6 inches wide and Type II reflective sheeting shall be a minimum of 6 inches wide.

Color fading. *The product shall show no appreciable fading after 70 hours of exposure in a weatherometer when tested in accordance with the following procedure:

Apply the material to three test panels 2.5 x 6 inches. Place the panels in a weatherometer (Atlas, Sunshine-type) fitted with an 18-102 (18 minutes of sunshine and rain, and 102 minutes of sunshine) cyclic gear. After 70 hours of exposure, visually compare the panels to unexposed standard panels for color fading.

Source: https://assist.dla.mil -- Downloaded: 202640-26T08:06Z
Check the source to verify that this is the current version before use.

*Color fading - An appreciable change in color means a change that is immediately noticeable in comparing the exposed specimen with the original comparison specimen. If a change in the angle of light is required to detect a slight change in color, the change is not appreciable.

Tensile strength. The product shall have a minimum tensile strength of 20 pounds per square inch at a temperature between 70° (21°C) and 80°F (26.7°C) with a rate of pull of 0.25 of an inch per minute using ASTM D 638.

Skid resistance. At 25°C, the product shall have a minimum skid resistance of 35 BPN when measured using ASTM E 303.

The issue of ASTM D 638 and ASTM E 303 in effect on the date of the solicitation shall be used to determine compliance with these requirements.

Certification. The contractor shall certify that the product offered meets the salient characteristics of this description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same product sold in the commercial marketplace. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements. The manufacturer shall utilize recovered materials to the maximum extent practicable.

Packaging, packing, and marking. The items shall be packaged in accordance with normal commercial practice and packed to assure acceptance by common carrier and provide product protection against loss and damage during multiple shipments, handling, and storage. The shipping container shall be in compliance with National Motor Freight Classification and Uniform Freight Classification. Shipping containers shall be marked as specified in the contract or order.

ASTM Standards are available from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

Standards for yellow and white highway colors are available from the Federal Highway Administration, Washington, DC 20590.

Note: In order to aid in the selection of size and type, the following additional information is provided:

The product accommodates common line widths (e.g. 4, 6, 8, 12, 16, and 24 inches), and words and symbols (e.g. "STOP", "ONLY", "SCHOOL", "R's", turn arrow, and straight arrow).

Class 1 - (All types) The product is intended to be imbedded into new asphalt concrete pavement during paving operations, but may also be applied to stable existing clean, dry asphalt or portland cement concrete pavement.

Class 2 - (All types) The product is intended only for application to stable, existing, clean dry asphalt and portland cement concrete pavements.

Application shall be in accordance with manufacturer's instructions.

Civil Agency Coordinating Activities

DOT-FHW, FIS
USDA - AFS

Preparing activity:

GSA-FSS

GSA DC-01105019

Source: https://assist.dla.mil -- Downloaded: 202610-26T08:06Z
Check the source to verify that this is the current version before use.